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DATA COMPUTER NETWORKS
Network Port and Cyber Security

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Cybersecurity has become one of the most critical fields in today's digital world, driven by the rapid growth of the internet and the widespread reliance on interconnected systems. As individuals, businesses, and governments increasingly depend on digital networks to store, process, and transmit information, the need to protect these systems from cyber threats has never been greater.

A fundamental element of network communication is the use of network ports, which act as gateways that allow different services and applications to exchange data. While these ports are essential for enabling connectivity and functionality, they can also become potential entry points for attackers if they are misconfigured, left open unnecessarily, or not properly secured. For this reason, understanding how network ports work and implementing effective security measures to protect them is a crucial part of maintaining a strong cybersecurity posture. [1],[3]

Network Port

Whenever any application in one computer sends data to another application of a different computer then it sends using IP Address and MAC Address but how does our computer know that this data is for a specific application and this data is sent by any specific application? There comes the concept of Port.

For instance, imagine your **MAC Address** or **IP Address** as the PIN code of the nearest Post Office and your house address as a Port. Whenever any parcel is sent to you it gets received by the nearest post office and then it is identified by your address where to deliver that parcel. Similarly in a computer data is first received using their IP or MAC address then it is delivered to the application whose port number is with the data packets.[1],[2]

Port is a logical address of a 16-bit unsigned integer that is allotted to every application on the computer that uses the internet to send or receive data.

Now every time any application sends any data, it is identified by the port that which the application sent that data and the data is to be transferred to the receiver application according to its port. We often call port as port number.

In the **OSI Model** ports are used in the Transport layer. In the headers of Transport layer protocols like **TCP and UDP**, we have a section to define port (port number). The network layer has to do nothing with ports, their protocols only care about IP Addresses.

Ports are assigned by computer i.e., operating system to different applications. Ports help computer to differentiate between incoming and outgoing traffic. Since the port is a 16-bit unsigned number it ranges from 0 to 65535.[1],[5],[6]

Types of Ports

Ports are further divided into three categories:

1. Well Known Port.
2. Registered Port.
3. Dynamic Port.

1. Well Known Port

- It is from the range 0 to 1023
- It is reserved for common and specifically used service
- It is used by some widely adopted protocols and services like HTTP(port 80), FTP (port 21), DNS (Port 53), SSH (port 22), etc.....

2. Registered Port

- It is from range 1024 to 49151.
- These are used by applications or services that are not as common.
- But it is used by those applications or services which require its specific port.
- Organizations can ask IANA (Internet Assigned Number Authority) for any specific port number within this range.

3. Dynamic Port

- It is from range 49152 to 65535.
- It is also known as Ephemeral or Private Port.
- It is used for those connections that are temporary or short-lived.
- It is not registered or assigned and can be used by any process.[2],[4]

Importance of Ports

Ports have many significances. There is some of them:

- **Identification of service-** Different application/services that work on the same device can be differentiated by their port numbers. For example, HTTP(Port number 80) and SMTP (port number 25) in the same computer uses different port number to ensure their data goes to the correct service.
- **Efficient Data Routing-** When a network device receives data from different places it uses port numbers to efficiently route those data packets to the respective application.
- **Block traffic from specific applications/services-** When we have to block incoming or outgoing traffic from a specific application/service then we need to install a firewall and specify the port number of that application/service. We block traffic from/to some specific applications/services when we find any potential threats from those applications/services.
- **Scalability of services-** Many services can run simultaneously on the same device and can be differentiated using their port number. This helps the device to scale and support many services at the same time.[1],[3]

Some Popular Port Numbers

Some common/Popular port numbers that are used by those applications/services which are frequently used by us:[4],[5],[6]

Port Number	Used By
80	HTTP (Hyper Text Transfer Protocol)
23	Telnet
25	SMTP (Simple Mail Transfer Protocol)
53	DNS (Domain Name System)
7	Echo
20/21	FTP (File Transfer Protocol)
69	TFTP (Trivial File Transfer Protocol)
443	HTTPS (Hyper Text Transfer Protocol Secure)
22	SSH (Secure Shell)
110	POP3(Post Office Protocol version 3)
67/68	DHCP (Dynamic Host Configuration Protocol)
123	NTP (Network Time Protocol)
143	IMAP (Internet Messaging Access Protocol)
1433	Microsoft SQL
3306	MySQL
5432	PostgreSQL
27017	MongoDB

Cybersecurity

Cybersecurity includes the tools and practices used to protect systems, networks, and data from online attacks. These attacks often aim to steal sensitive information like personal or financial details. As cyber threats keep evolving, strong cybersecurity is essential to protect privacy, maintain a company's reputation, keep operations running smoothly, and reduce the financial damage caused by security breaches.[3],[8]

Cyber threats associated with ports

1. Port Scanning

Port scanning is a technique attackers use to identify which ports on a device or server are open. Open ports often reveal which services are running (web servers, remote access, databases, etc.). If a vulnerable or misconfigured service is found, an attacker may use it as an entry point into the system.

2. Brute Force Attacks

A brute force attack happens when an attacker repeatedly tries different password combinations until one works. This is usually done using automated tools that can test thousands of passwords quickly. Weak, common, or reused passwords make these attacks much more likely to succeed.

3. Malware

Malware is harmful software designed to damage systems, steal information, or gain unauthorized control. It includes viruses, worms, ransomware, spyware, and trojans, each with different objectives. Malware often spreads through infected downloads, email attachments, fake software, or compromised websites.

4. Phishing

Phishing is a social engineering attack where scammers pretend to be a trusted person or organization. They use fake emails, messages, or websites to trick users into entering passwords, bank details, or other sensitive data. It's very common because it targets human mistakes rather than technical weaknesses.

5. DDoS Attacks

A DDoS (Distributed Denial of Service) attack floods a server or website with a large amount of traffic. This overload can slow down services, crash systems, or make them unavailable to legitimate users. Attackers often use botnets (many infected devices) to generate the traffic.

6. Man-in-the-Middle (MITM) Attacks

MITM attacks occur when an attacker intercepts communication between two systems without the users knowing. This can allow them to steal data, modify information, or inject malicious content into the connection. For example, if users access websites over unencrypted HTTP (port 80) on an unsafe network, an attacker may capture credentials or alter traffic.

7. Port Spoofing & Redirection

Attackers may run a fake service on a port to imitate a legitimate one and trick users or systems into connecting. This can redirect victims to spoofed login pages or malicious servers to steal credentials. Misconfigured firewall rules or exposed services can make this type of deception easier to carry out.

8. Advanced Persistent Threats (APTs) using Ports

In targeted attacks, APT actors may abuse exposed ports to gain access and maintain long-term control over a network. They often establish command-and-control (C2) channels using common ports (like 443) or unusual ports to avoid detection. Over time, they may silently monitor activity, move laterally, and steal sensitive data.[1],[3],[8]

Cybersecurity Defense Mechanisms

1. Firewalls

A firewall is a security control that helps protect a device or network by controlling inbound and outbound traffic. For example, Windows Defender Firewall can block or allow connections based on rules such as the source/destination IP address, protocol (TCP/UDP), and port number. It can also apply rules based on specific programs and services, which helps ensure that only trusted applications are allowed to communicate over the network and reduces the risk of unauthorized access.

2. Intrusion Detection System (IDS)

An Intrusion Detection System (IDS) is a hardware device or software tool that continuously monitors network or system activity to detect suspicious behavior, unauthorized access, or potential attacks. Its main role is to alert administrators and generate detailed logs to support investigation and incident response.

How it works:

- **Monitoring and analysis:** The IDS observes network traffic or host activity and looks for unusual behavior.
- **Pattern/rule matching:** It compares activity against known attack signatures and rules (or checks for anomalies).

- **Alerts and logging:** When something suspicious is detected, it triggers an alert and records logs for further analysis.

3. Intrusion Prevention System (IPS)

An Intrusion Prevention System (IPS) is similar to an IDS, but it goes a step further. Instead of only detecting threats, an IPS is usually placed in line with network traffic so it can actively stop attacks in real time.

What an IPS can do:

- Block or drop malicious traffic before it reaches the target system.
 - Reset connections (for example, terminating suspicious sessions).
 - Prevent exploitation attempts by enforcing security rules automatically.
- Because IPS takes action automatically, it often requires careful tuning to reduce false positives and avoid blocking legitimate traffic.

4. Cryptography

Cryptography uses mathematical methods to protect information and communication from unauthorized access. A cryptographic protocol is a set of rules that uses cryptographic algorithms to achieve key security goals during communication and data exchange.

These protocols help provide:

- Confidentiality: keeping data private through encryption.
- Integrity: ensuring data is not modified (often using hashing).
- Authenticity: verifying identities (often using certificates/digital signatures).

Examples of cryptographic protocols include TLS (HTTPS), SSH, and IPsec, which are widely used to secure modern digital communications.[3],[7]

Case Studies

1.WannaCry Ransomware – A Single Open Port That Crippled the World

The WannaCry ransomware attack of 2017 demonstrated how a single exposed network port can lead to global disruption. The attack exploited Port 445, used by the Windows Server Message Block (SMB) protocol, which had been left open and unpatched on thousands of systems. By abusing the EternalBlue Vulnerability, attackers were able to execute malicious code remotely without authentication, gaining full control of vulnerable machines.

Once inside a system, WannaCry encrypted critical files and rapidly propagated across internal networks by scanning for other devices with same open port. This lateral movement allowed the malware to spread automatically, disabling hospitals, government institutions, and businesses worldwide. The incident proved that open and poorly managed ports can serve as direct attack vectors, turning routine network services into channels for large-scale cyber disasters.

2. Target Data Breach – Lateral Movement Through Trusted Ports

The 2013 Target data breach revealed how weak internal port control can undermine even large, well-funded organization. Attackers initially accessed Target's network through a third-party HVAC contractor whose systems were connected via insufficiently restricted network ports. Because the internal network lacked proper segmentation, open ports allowed attackers to move laterally from low-security systems to critical payment infrastructure.

By exploiting trusted internal connections, the attackers reached Point-of-Sale servers and collected millions of customer credit card records. The stolen data was then exfiltrated through legitimate ports such as Port 443, making the attack difficult to detect. This breach highlighted that failure to isolate systems and tightly control internal ports enables attackers to transform minor access points into pathways for massive data theft. [3],[9]

AI in Network Port Protection and Cybersecurity

Artificial Intelligence (AI) enables faster and more accurate detection by analyzing massive volumes of network traffic in real time. Unlike manual or static systems, AI continuously learns and adapts to new attack patterns.

AI & Machine Learning Techniques Used

- **Random Forest:** Random Forest classifiers are widely used in port-based cybersecurity due to their high accuracy and robustness. They operate by combining multiple decision trees to classify network traffic as benign or malicious. In port protection systems, Random Forest models analyze features such as destination port numbers, packet size distributions, source IP reputation, and traffic frequency. According to documented research, these models achieve 95–98% accuracy in detecting port scanning and unauthorized access attempts. Their ensemble nature reduces overfitting and improves reliability, making them suitable for large-scale port network environments where consistent performance is critical.[10],[11]
- **Support Vector Machines (SVM):** Support Vector Machines are particularly effective for real-time anomaly detection in high-dimensional network data. SVM models create decision boundaries that separate normal traffic patterns from abnormal or malicious behavior. In port cybersecurity applications, SVMs detect subtle deviations from baseline traffic, even when attack signatures are unknown. The document highlights that SVMs achieve 92–96% detection accuracy while maintaining low computational overhead, allowing deployment at network edges where rapid response and minimal latency are essential.

- **Deep Neural Networks (DNN):** Deep Neural Networks represent a major advancement in AI-driven cybersecurity. Unlike traditional machine learning models, DNNs do not rely on manual feature engineering. Instead, they automatically extract hierarchical features directly from raw packet-level data. This capability enables them to identify complex and sophisticated port-based attacks, including zero-day exploits with no historical attack data. Research cited in the document reports 94–99% accuracy in detecting advanced threats, making DNNs highly effective for protecting critical port infrastructure against evolving cyber risks.
- **Long Short-Term Memory (LSTM) Networks:** LSTM networks are a specialized form of recurrent neural networks designed to capture temporal patterns in sequential data. Port-based cyberattacks often follow a sequence: reconnaissance, port scanning, exploitation, and escalation. LSTM models analyze traffic behavior over time to recognize these attack phases and predict future malicious actions. The document indicates that LSTM-based systems achieve 91–97% accuracy in predicting attack progression, enabling proactive defense measures that stop attacks before they cause operational damage.

In real systems, these models are combined to improve reliability and reduce false alarms.[10],[11],[12]

Benefits of AI-Based Port Security

- Faster detection and response (milliseconds instead of minutes).
- Higher detection accuracy and fewer false alarms.
- Reduced financial losses and operational downtime.
- Improved protection for critical port and logistics infrastructure.

Automated Response and Resilience

AI-based cybersecurity systems enable automatic and immediate response to port-based attacks. When malicious activity is detected, the system can block source IPs, close vulnerable ports, and isolate affected systems within milliseconds.

Unlike traditional security solutions that require manual intervention and take minutes to respond, AI systems react in under one second, significantly reducing damage. Continuous monitoring and adaptive learning improve system resilience, allowing faster recovery and stronger protection against future attacks.

This automated response is essential for port infrastructure, where downtime can disrupt logistics, trade, and critical operations.

Challenges and Considerations

Despite the advantages of AI-based cybersecurity, several challenges remain. AI models can be difficult to interpret, making it hard to explain why certain traffic is classified as malicious. Attackers may also attempt to deceive or manipulate AI systems using specially crafted network traffic. In addition, deploying AI security solutions requires high-quality data, proper training, and skilled personnel. These factors must be carefully considered when implementing AI-driven security in port infrastructure.

Future of AI in Port Cybersecurity

The future of port cybersecurity relies on more autonomous and intelligent AI systems. Advanced AI will enable predictive threat detection, faster automated responses, and improved protection of smart ports and connected maritime systems. Collaborative approaches such as federated learning will allow ports to strengthen security without sharing sensitive data. As cyber threats continue to evolve, AI will become an essential component of resilient and secure port infrastructure. [3],[8],[11]

Conclusion

In today's highly connected digital environment, cybersecurity has become a fundamental requirement for protecting systems, data, and critical infrastructure. Network ports play a vital role in enabling communication between applications and services, but they also represent significant security risks when improperly managed. As demonstrated throughout this project, open or misconfigured ports can be exploited through various cyber threats such as port scanning, malware, brute force attacks, DDoS attacks, and advanced persistent threats.

Traditional security mechanisms like firewalls, IDS, and IPS provide essential protection, but they are often limited by static rules and slower response times. The integration of Artificial Intelligence into port cybersecurity introduces a more adaptive and proactive defense approach. AI and machine learning techniques including Random Forest, SVM, Deep Neural Networks, and LSTM enable accurate detection of known and unknown threats, real-time analysis of network traffic, and prediction of attack behavior.

Furthermore, AI-based automated response mechanisms significantly enhance resilience by blocking malicious activity, isolating affected systems, and reducing response times from minutes to milliseconds. Although challenges such as model interpretability, data quality, and skilled workforce requirements remain, the benefits of AI-driven port security far outweigh these limitations.

In conclusion, securing network ports using advanced cybersecurity techniques and Artificial Intelligence is essential for ensuring the reliability, safety, and continuity of modern digital and port infrastructures. As cyber threats continue to evolve, AI-powered cybersecurity solutions will play a critical role in building resilient, secure, and future-ready networks.

References

- [1] J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach, 8th ed. Pearson, 2021.
- [2] A. S. Tanenbaum and D. J. Wetherall, Computer Networks, 6th ed. Pearson, 2021.
- [3] W. Stallings, Network Security Essentials: Applications and Standards, 6th ed. Pearson, 2020.
- [4] Internet Assigned Numbers Authority (IANA), “Service Name and Transport Protocol Port Number Registry,” 2024.
- [5] IETF, “RFC 793: Transmission Control Protocol (TCP),” 1981.
- [6] IETF, “RFC 768: User Datagram Protocol (UDP),” 1980.
- [7] K. Scarfone and P. Mell, Guide to Intrusion Detection and Prevention Systems (IDS/IPS), NIST SP 800-94, 2012.
- [8] Verizon, Data Breach Investigations Report (DBIR), 2023.
- [9] Symantec Corporation, WannaCry Ransomware Attack Analysis, 2017.
- [10] A. L. Buczak and E. Guven, “A survey of data mining and machine learning methods for cyber security intrusion detection,” IEEE Communications Surveys & Tutorials, vol. 18, no. 2, pp. 1153–1176, 2016.
- [11] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.
- [12] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.